

7 Files and streams

Before you can read from a file, you must open it. After you are done reading from a file, you must close it. There are two common varieties of reading:

- reading characters (a character is 16 bits long)
- reading bytes (a byte is 8 bits long)

Inevitably, when you sit down to read from a file, you have to sort through the choices on those lists. The best approach is to pick one from either list—character and byte—and learn to use it.

7.1 Reading Characters from a File

Say you want to read from a file:

- You will need to open the file.
- You will need to read from the file to the end.
- You will need to close the file.

First of all, what is a file? A file is an instance of the class `File`

```
import java.io;

public class FileRead
{
    public FileRead()
    {
        File inFile = new File( "C:/orig/aFile.txt" );
    }
    public static void main( String[] args )
    {
        FileRead fr = new FileRead();
    }
}
```

All the classes used in I/O come from this package. Because the constructor on `File` throws an `IOException`, we are forced to place it in a try-catch block:

```
import java.io;

public class FileRead
{
    public FileRead()
    {
        try
```